

SAFETY REPORT: TR-SAF-2026-902

Product: BCU-800V Safe State & Pyro-Switch Subsystem | ISO 26262 ASIL D

1. Contactor De-Energization Latency (REQ-BCU-007):

Emergency contactor trip latency measured at 6.40 ms under simulated overvoltage event (threshold ≤ 10.0 ms). Dual-channel high-side and low-side driver de-energized simultaneously without contact welding.

2. Pyro-Fuse Ignition Timing (REQ-BCU-010):

Hardware short-circuit analog comparator triggered pyro-switch in 2.8 microseconds. Total fault current interruption accomplished in 1.1 ms.

3. Pressure Relief & Gas Venting Dynamics (REQ-BCU-013):

Computational fluid dynamics simulation and gas venting rate calculation completed for 300 mbar burst target; physical burst validation fixture pending.

4. Acoustic Leakage Impedance Analysis (REQ-BCU-029):

Acoustic impedance sensor characterization demonstrated signal dampening in presence of liquid solvent, but exact dB threshold calibration requires vehicle pack installation test.

5. Firing Circuit Capacitive Energy (REQ-BCU-030):

Firing capacitor energy calculation confirmed 4.8 J nominal theoretical capacity; physical discharge waveform capture pending final harness qualification.